

Human Impact Section

Interactive Implementation Guide

Detailed implementation guide for the interactive Human Impact section of the Hawaiian Coot conservation website.

The Human Impact section should behave like an interactive ecosystem simulation. Users should visually experience the difference between environmental destruction and habitat recovery. The page is designed to:

- demonstrate cause-and-effect relationships
- emotionally engage users
- visually connect human actions to wetland health
- encourage conservation participation

1. Core Interactive Concept

The page is divided into two environmental states:

LEFT SIDE → Harm
RIGHT SIDE → Hope

The center ecosystem reacts dynamically depending on user interaction.

Hovering over:

- Harm cards → ecosystem deteriorates
- Hope cards → ecosystem recovers

2. Central Ecosystem Visualization

The center panel is the main storytelling element.

Design:

- Split-screen ecosystem
- Damaged wetlands on left
- Healthy wetlands on right
- Hawaiian Coot positioned near center

Interactive Slider:

- User drags slider horizontally
- Ecosystem transitions in real time
- Colors shift dynamically
- Bird behavior changes

3. Harm Side Interactions

Wetland Drainage:

- Water level lowers
- Cracked mud appears
- Vegetation dries out
- Red warning glow intensifies

Introduced Predators:

- Predator silhouettes appear
- Nesting icons flash red
- Birds scatter

Pollution & Disturbance:

- Water becomes murky
- Floating pollution particles appear
- Habitat darkens gradually

4. Hope Side Interactions

Refuge Creation:

- Wetlands brighten
- Water ripples return
- Habitat glow appears

Taro Farmer Partnerships:

- Taro vegetation grows
- Water irrigation flows
- Habitat health increases

Predator Control:

- Safety shield appears
- Nesting success rises
- Chick icons appear

5. Ecosystem Health Meter

Add live ecosystem indicators below the center visualization.

Suggested Indicators:

- Wetland Health
- Nesting Success
- Water Quality
- Population Stability

Values update dynamically based on interactions.

6. Hover Animation System

Every card triggers a specific ecosystem reaction.

Hover Effects:

- Glow intensifies
- Background subtly animates
- Ecosystem metrics update
- Tooltips fade into view

Additional Effects:

- Particle systems
- Water ripple effects
- Dust/smoke particles

7. How You Can Help Section

Each action card should trigger a positive ecosystem response.

Support Wildlife Refuges:

- Protected areas glow
- Habitat health rises

Manage Pets:

- Nesting success improves

Wetland Restoration:

- Plants regrow
- Water levels rise

8. Recommended Technologies

Recommended Libraries:

- React
- Framer Motion
- GSAP
- CSS keyframes

Recommended Features:

- Shared global state
- Smooth crossfade transitions
- Dynamic lighting changes
- Bird movement animations

9. Interaction Trigger Table

USER ACTION	VISUAL CHANGE	ANIMATION	MESSAGE
Hover Wetland Drainage	Wetlands dry	Heat shimmer + cracks	Habitat loss destroys nesting grounds
Hover Introduced Predators	Birds scatter	Predator silhouette fade-in	Predators threaten chicks
Hover Pollution	Water darkens	Murky ripple effect	Pollution damages food sources
Hover Refuge Creation	Wetlands brighten	Ripple + glow	Protected habitats improve survival
Hover Taro Partnerships	Vegetation grows	Growth animation	Traditional agriculture supports wetlands
Hover Predator Control	Nesting success rises	Shield animation	Protection improves survival
Move Ecosystem Slider	Scene transforms	Smooth crossfade	Human choices shape habitat future

10. Final Design Philosophy

The Human Impact section should feel like an interactive conservation documentary. Every animation should visually reinforce the connection between human activity, habitat health, wetland conservation, and Hawaiian Coot survival.